

OCR TELUGU TEXT EXTRACTION MODEL

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Abstract

In today's digital age, the preservation and accessibility of historical, cultural, and linguistic content are of paramount importance. Many valuable documents, particularly those in regional languages such as Telugu, remain confined to physical formats, including books, manuscripts, and handwritten notes. These physical documents are susceptible to deterioration, loss, and limited accessibility, posing significant challenges for their preservation and utilization. The proposed solution involves the use of advanced Optical Character Recognition (OCR) techniques to accurately extract text from uploaded Telugu PDFs, providing users with an editable text display to review and correct the extracted content.

Aim of the project

The aim of this project is to develop a system that accurately extracts and digitizes text from Telugu PDF documents using advanced Optical Character Recognition (OCR) techniques, thereby preserving and enhancing the accessibility of historical, cultural, and linguistic content. By converting these physical documents into digital format, we can ensure their preservation, improve accessibility, and facilitate the use of this content by a wider audience.

Objectives

- Develop a user-friendly interface for uploading Telugu PDF documents.
- Utilize advanced Optical Character Recognition (OCR) technologies to accurately extract text from Telugu PDF documents. This involves converting scanned images of text into machine-encoded text.
- Provide an editable text display for users to review and correct the extracted text. Allow users to flag specific sections of text that may need further verification or reprocessing.
- Provide functionality for users to save the edited and corrected text in a structured format (JSON) and allow them to download it for future reference or use.
- Digitize Telugu PDF documents to preserve valuable historical, cultural, and linguistic content, and make it accessible to a broader audience, including researchers, historians, linguists, and the general public.

Implementation

- Step 1:** Setting Up the Environment and Install Required Libraries like poppler-utils, opencv, lumpy, pdf2image, pytesseract, libtesseract, torch, transformers, gradio
- Step 2:** User will upload the pdf file in the interface, then using pdf2image library to convert the PDF pages to images.
- Step 3:** Preprocessing Images like rescaling frame size, applying gamma(Gamma correction is a technique used to adjust the brightness and contrast of an image), Adaptive thresholding (It is a binarization technique that converts a grayscale image into a binary image (black and white) based on the local properties of the pixels), Edge detection (It is a technique used to identify the boundaries of objects within an image), Morphological transformations (This is a image processing operation that process images based on their shapes. They usually apply to binary images)
- Step 4:** Extracting Text Using OCR: Using Tesseract to extract text from the preprocessed images.

```
# Function to extract text from processed image
def extract_text_from_image(image, lang='tel'):
    return pytesseract.image_to_string(image, lang=lang)
```

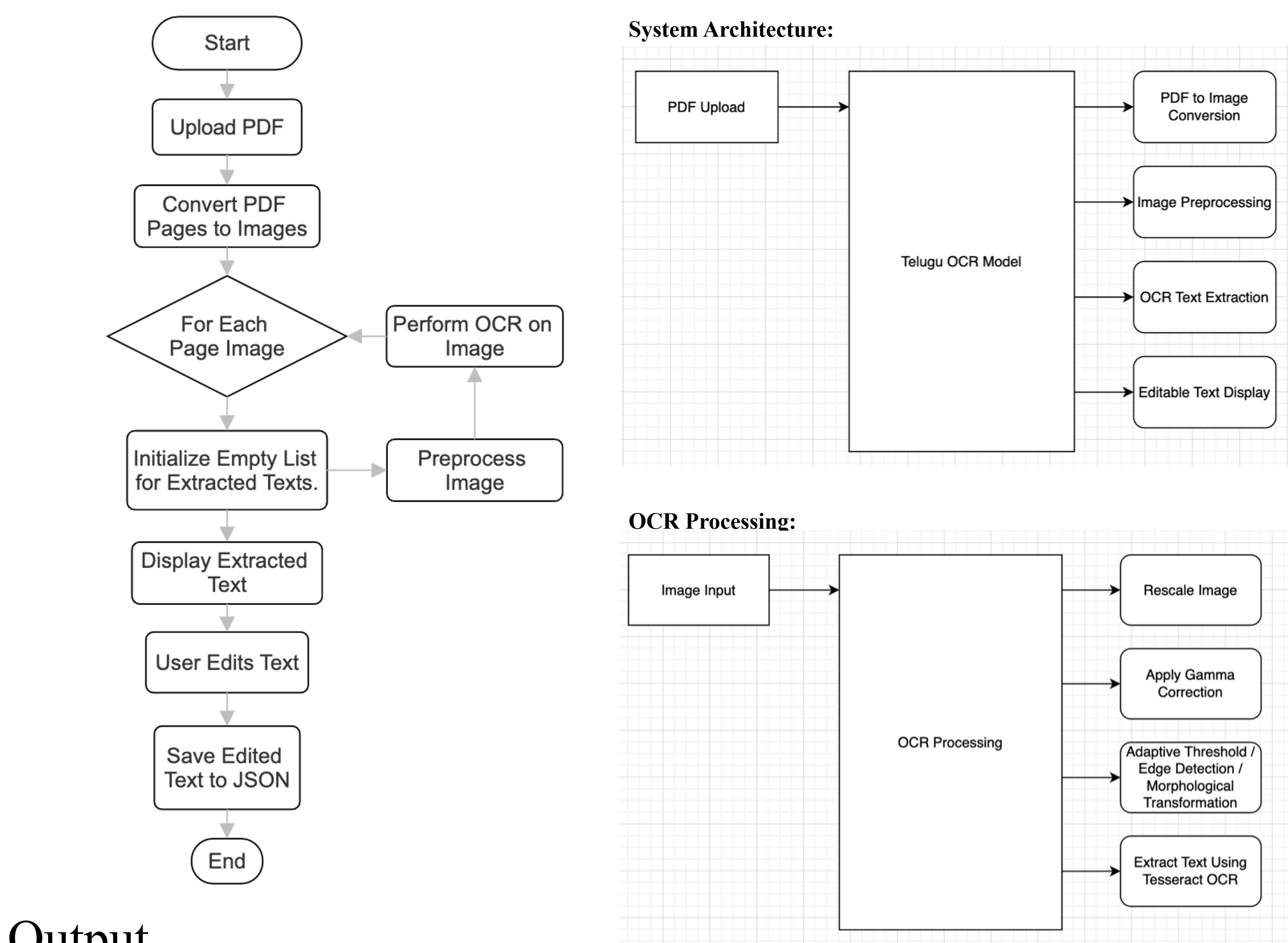
Step 5: Creating a Gradio interface for uploading the PDF and selecting the preprocessing method to get the extracted text as the output.

```
interface = gr.Interface(
    fn=process_pdf,
    inputs=[
        gr.File(label='Upload PDF', file_types=['pdf']),
        gr.Dropdown(choices=methods, label='Preprocessing Method'),
    ],
    outputs=gr.Textbox(label="Extracted Text"),
    live=True,
)

# Launch the interface
interface.launch()
```

Step 6: Translating the Telugu text to English using MBartForConditionalGeneration, MBart50TokenizerFast (mBART is a multilingual sequence-to-sequence model designed for neural machine translation (NMT) and other sequence generation tasks. It is an extension of BART (Bidirectional and Auto-Regressive Transformers) that has been trained on multiple languages.)

Flow Chart and Block Diagrams

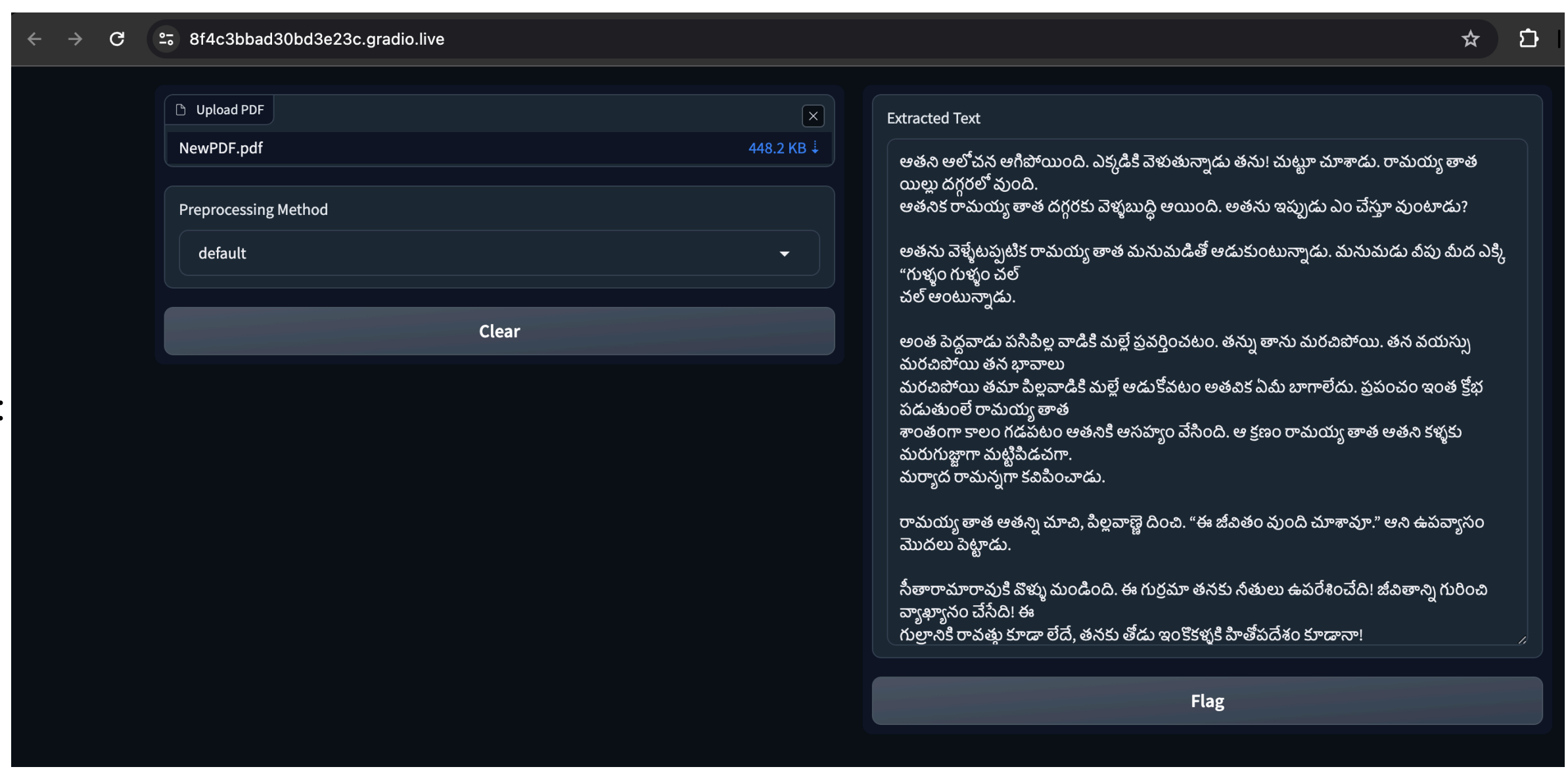


Output

Code output:



Gradio Interface:



Conclusion

we have developed a comprehensive system that facilitates the digitization of Telugu documents. This system not only extracts text from PDF files with high accuracy but also provides users with an interface to review, edit, and translate the extracted text into English. The project's multifaceted approach addresses the critical need for digital preservation of regional languages and enhances the usability of this content for a broader audience, including researchers, historians, linguists, and the general public.

Impact and Future Scope

The system could be expanded to support the extraction of text from documents containing multiple languages, often seen in historical Telugu documents. Integrating Handwritten Text Recognition (HWR) capabilities would enable the processing of handwritten Telugu manuscripts and notes, unlocking a vast trove of historical information.

References

- **Data Set:** <https://github.com/tesseract-ocr/tessdata/blob/main/script/Telugu.traineddata>
- **OCR FOR TELUGU LANGUAGE USING TESSERACT:** <https://www.ijfans.org/uploads/paper/15ef8a7470e502f61dd77d390899bec1.pdf>
- **Tesseract OCR Software Tutorial:** <https://guides.nyu.edu/tesseract>
- **OCR for Printed Telugu Documents:** <https://people.iith.ac.in/saketha/research/rajeshMTP.pdf>
- **IEEE Dataport:** <https://ieee-dataport.org/documents/ocr-telugu-image-dataset>
- **Optical Character Recognition:** <https://www.techtarget.com/searchcontentmanagement/definition/OCR-optical-character-recognition>
- **Tesseract OCR:** <https://github.com/tesseract-ocr/tesseract>
- **Tesseract documentation:** <https://tesseract-ocr.github.io/>